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Design and Implementation of an FSM-Based Four-Quadrant DC Motor Drive Using ESP32Presented

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Abstract

This project presents the design and implementation of an FSM-based DC motor drive system using the ESP32 microcontroller. The system controls the motor speed and direction using Pulse Width Modulation (PWM) and an H-Bridge converter. Different operating modes, including forward run, reverse run, dynamic braking, and stop modes, are achieved through appropriate switching sequences.

A Finite State Machine (FSM) is used to ensure safe and reliable transition between operating states. Dead time is implemented during switching transitions to prevent shoot-through faults and protect the power switching devices.

The hardware design utilizes TLP250 gate drivers for MOSFET isolation and gate driving. The system was modeled and verified using MATLAB/Simulink and Proteus prior to hardware implementation and PCB design.

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Introduction

DC motors are widely used in industrial, robotics, and automation applications due to their simple structure, high efficiency, and ease of speed control. DC motor drives play an important role in controlling motor operation, including speed regulation, direction control, and braking functions.

In this project, an ESP32-based DC motor drive system is designed and implemented using a MOSFET H-Bridge configuration and TLP250 gate drivers. A Finite State Machine (FSM) is employed to manage different operating modes, including forward rotation, reverse rotation, forward braking, reverse braking, and coasting operation.

Pulse Width Modulation (PWM) is utilized to achieve efficient motor speed control. In addition, dead time is introduced during switching transitions to prevent shoot-through faults and ensure safe operation of the power switches.

The proposed system is verified through simulation using MATLAB/Simulink and Proteus before hardware implementation and PCB design.

Objectives of the Project

- Control DC motor speed using Pulse Width Modulation (PWM).
- Control motor direction in forward and reverse modes.
- Implement dynamic braking modes.
- Design and implement a MOSFET H-Bridge drive circuit.
- Interface the ESP32 microcontroller with the power stage.
- Use TLP250 gate drivers for MOSFET isolation and gate driving.
- Implement safe switching using dead time protection.
- Develop a Finite State Machine (FSM) for motor state control.
- Simulate and verify the system using MATLAB/Simulink and Proteus.
- Design a PCB for hardware implementation.

Components Used

Component	Image	Function
ESP32		Main controller that manages the motor drive system and generates PWM control signals
MOSFET IRF540N		Performs high-speed power switching in the H-Bridge circuit
TLP250		Provides MOSFET gate driving and electrical isolation
DC Motor		Mechanical load driven by the H-Bridge circuit
Power Supply		Provides electrical power for the control and power circuits

System Overview

The DC motor drive system is designed to control the speed and direction of a DC motor using an ESP32 microcontroller, a MOSFET-based H-Bridge power stage, and TLP250 gate driver circuits.

The system enables forward and reverse rotation, as well as braking and coasting operations, while regulating motor speed using Pulse Width Modulation (PWM). A Finite State Machine (FSM) is implemented in the ESP32 to manage the different operating modes and ensure safe transitions between states.

The ESP32 receives user inputs through push buttons for direction and mode selection, while a potentiometer is used to set the desired speed reference. Based on these inputs, the microcontroller generates PWM signals for speed control and digital control signals for direction and mode selection.

The PWM signal controls the motor speed by varying its duty cycle. Increasing the duty cycle increases the average voltage applied to the motor, resulting in higher speed, while decreasing the duty cycle reduces the motor speed.

The generated control signals are sent to the TLP250 gate driver circuits, which provide electrical isolation between the low-voltage control side and the high-power stage, as well as sufficient gate drive current for fast switching of the MOSFETs.

The MOSFETs are configured in an H-Bridge topology, which enables bidirectional current flow through the motor. By appropriately switching the MOSFET pairs, the system achieves forward rotation, reverse rotation, dynamic braking, and coasting modes.

Finally, the DC motor responds to the switching states and PWM signals generated by the ESP32, allowing full control over motor speed, direction, braking, and stopping behavior.

DC Motor Drive System Operating Concept

The operating concept of the DC motor drive system starts with user inputs connected to the ESP32 microcontroller. The user selects the desired operating mode, including Forward Run, Reverse Run, Braking, or Stop, while the motor speed is controlled using a potentiometer.

The ESP32 continuously reads all input signals. Push buttons are used to determine the required motor direction and operating mode, while the potentiometer provides an analog voltage proportional to the desired motor speed.

The analog signal from the potentiometer is converted into a digital value using the ESP32's Analog-to-Digital Converter (ADC). Based on this value, the microcontroller generates a Pulse Width Modulation (PWM) signal to control the motor speed.

The PWM signal determines the average voltage applied to the motor:

• A higher duty cycle results in a higher average voltage and increased motor speed.

• A lower duty cycle results in a reduced average voltage and lower motor speed.

However, the ESP32 cannot directly drive the MOSFET gates due to limited voltage and current capability. Therefore, the control signals are sent to TLP250 gate driver circuits.

The TLP250 provides electrical isolation between the low-power control side and the high-power motor drive stage using an internal optocoupler. This protects the ESP32 from voltage spikes, switching noise, and electrical disturbances generated by the motor drive circuit.

In addition, the TLP250 amplifies the control signals and provides sufficient gate current to ensure fast and reliable switching of the MOSFETs.

The outputs of the gate drivers are connected to a MOSFET-based H-Bridge configuration, which controls the direction of current flow through the DC motor.

Depending on the switching combination:

• **Forward rotation** is achieved by allowing current flow in one direction.

• **Reverse rotation** is achieved by reversing the current direction.

• **Braking (Plugging)** is achieved by applying reverse voltage across the motor terminals, producing a strong opposing torque that rapidly reduces motor speed.

• In the OFF state, all MOSFETs are disabled, and the motor enters **coasting mode**.

During operation, the ESP32 continuously updates the PWM duty cycle and switching states according to user commands. This enables complete electronic control of motor speed, direction, and braking in a safe and efficient manner.

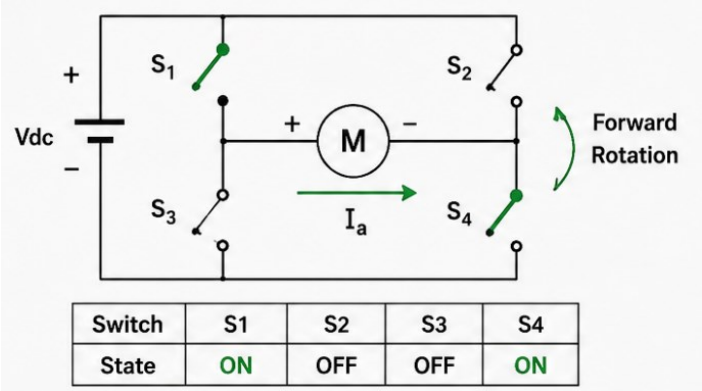
H-Bridge Operating Principle

The H-Bridge circuit is used to control the direction and operating modes of a DC motor by controlling the switching states of four MOSFETs (Q1, Q2, Q3, and Q4). The ESP32 generates the control signals based on the selected operating mode, while the MOSFET configuration determines the polarity of the voltage applied to the motor terminals.

The DC motor is connected between the two central nodes of the H-Bridge. By activating specific MOSFET pairs, the current direction through the motor is controlled, enabling forward rotation, reverse rotation, braking, and coasting operations.

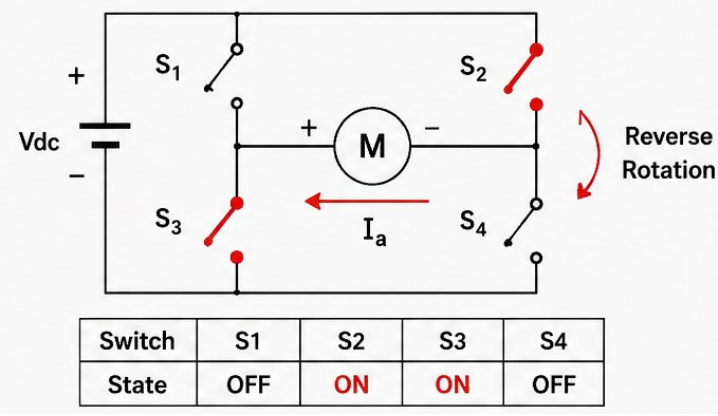
Operating Modes:

• **Forward Run (F.R):** Q1 and Q4 are ON. Current flows through the motor in one direction, resulting in clockwise rotation.



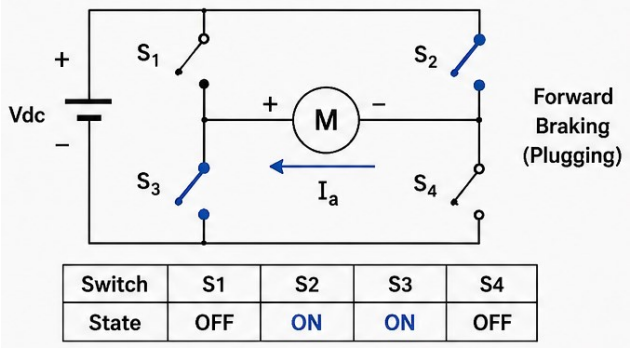
Switch	S1	S2	S3	S4
State	ON	OFF	OFF	ON

• **Reverse Run (R.R):** Q2 and Q3 are ON. The current direction is reversed, causing counterclockwise rotation.



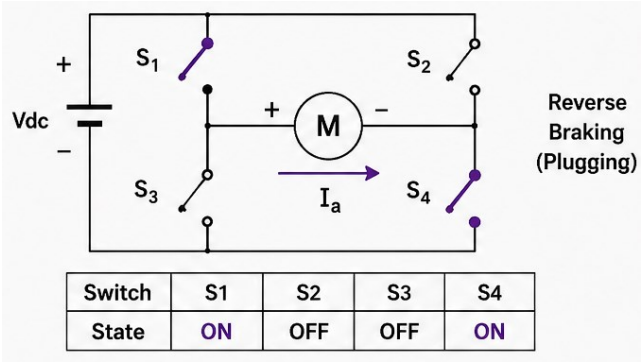
Switch	S1	S2	S3	S4
State	OFF	ON	ON	OFF

• **Forward Braking (F.B):** Specific switching creates a braking condition that produces an opposing torque to reduce motor speed rapidly.



Switch	S1	S2	S3	S4
State	OFF	ON	ON	OFF

• **Reverse Braking (R.B):** Reverse switching configuration generates braking torque to decelerate the motor when rotating in reverse.



Switch	S1	S2	S3	S4
State	ON	OFF	OFF	ON

• **Coasting Mode:** All MOSFETs are OFF, and the motor terminals are floating, allowing the motor to decelerate naturally due to inertia and friction.

• **Stop Mode:** All MOSFETs are OFF, and no voltage is applied to the motor. The motor remains stationary.

PWM Speed Control

The speed of the DC motor is controlled using Pulse Width Modulation (PWM) generated by the ESP32 microcontroller. The duty cycle of the PWM signal determines the average voltage applied to the motor terminals, and consequently controls the motor speed.

A higher duty cycle results in a higher average voltage, leading to an increase in motor speed, while a lower duty cycle reduces the average voltage and decreases the motor speed.

The duty cycle is defined as:

$$\textbf{Duty Cycle} = \frac{Ton}{T} \times 100\%$$

Where:

Ton is the ON time of the PWM signal,

T is the total period of the PWM signal.

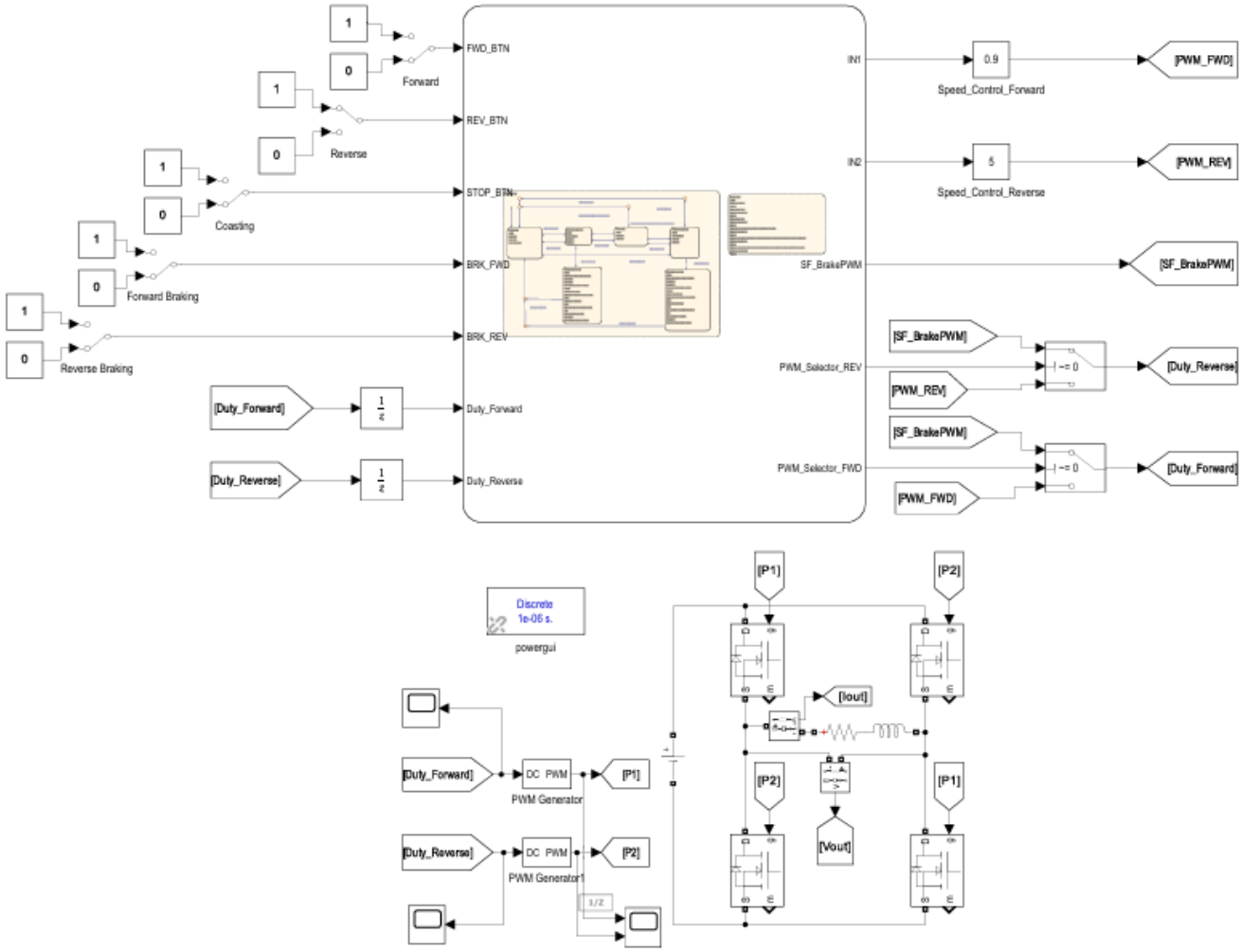
A potentiometer connected to the ESP32 is used as an analog input to vary the duty cycle, allowing smooth and continuous control of the motor speed.

Simulation (MATLAB/Simulink)

Simulation was carried out using MATLAB/Simulink to verify the proposed DC motor drive system before hardware implementation. The model includes a DC motor, H-Bridge converter, load resistance, and motor inductance to represent a realistic electrical behavior.

The H-Bridge was used to control the direction of current flow through the motor, enabling forward and reverse operation. PWM signals generated from the control system were applied to regulate the motor speed.

The simulation allowed observation of key electrical variables such as armature voltage, current response, and motor speed under different operating conditions including forward rotation, reverse rotation, and braking modes. This helped in validating the control strategy and ensuring correct system behavior before moving to the hardware stage.

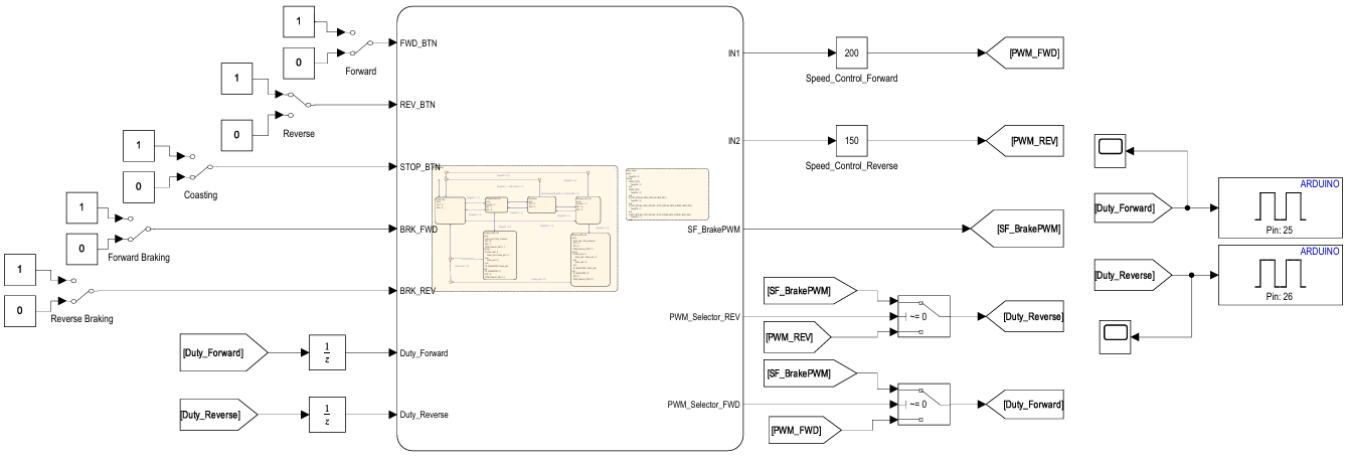


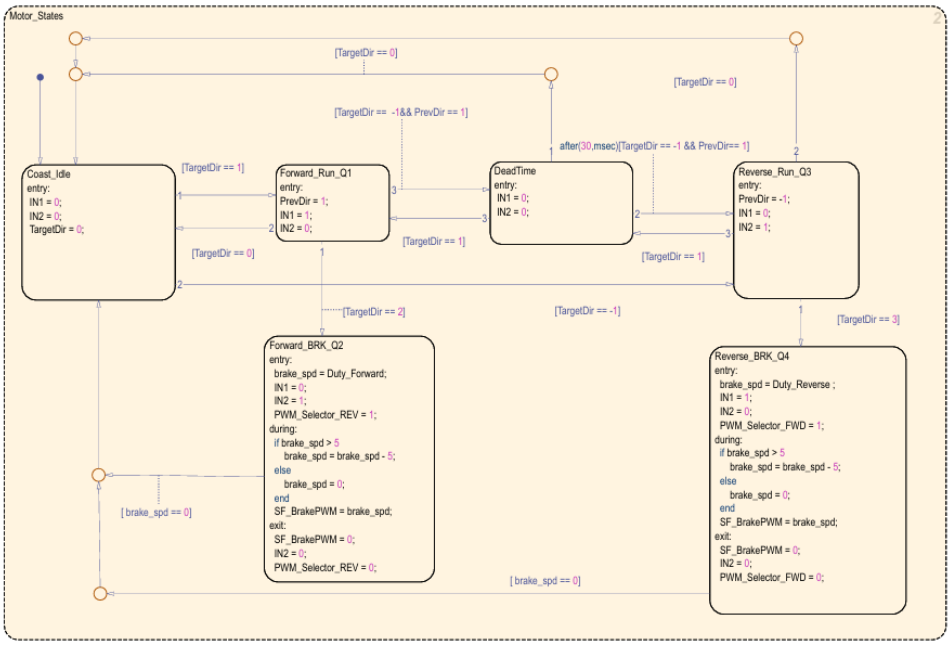
Hardware Target Model (ESP32 Control System)

The hardware control system is based on the ESP32 microcontroller, which is responsible for generating PWM signals and controlling the operating states of the DC motor drive system. The control logic is implemented using a Finite State Machine (FSM) to manage different operating modes such as forward, reverse, braking, and stop.

The ESP32 interfaces with the power stage through TLP250 gate drivers to ensure electrical isolation and proper gate driving for the MOSFETs in the H-Bridge circuit. Control signals (IN1, IN2, and PWM selectors) are generated according to the system state and user inputs.

This hardware model ensures safe, reliable, and flexible control of the motor drive system while protecting the microcontroller from high-power switching noise.





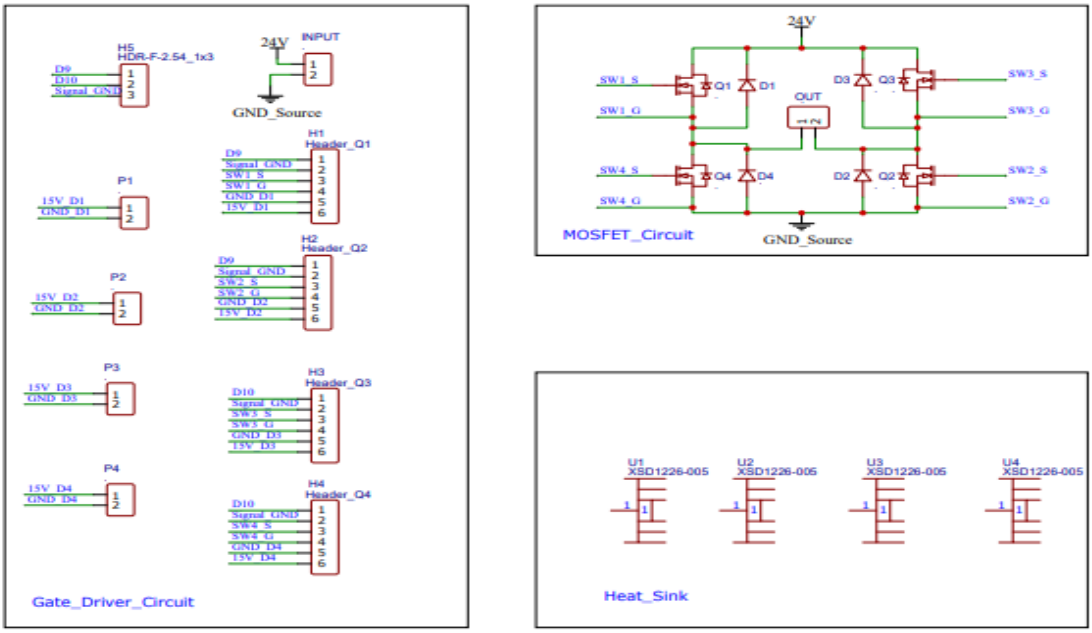

```
Input_Logic
entry:
  TargetDir = 0;
during:
  if(BRK_FWD)
    TargetDir = 2;
  end;
  if(BRK_REV)
    TargetDir = 3;
  end;
  if(STOP_BTN && ~BRK_FWD && ~BRK_REV)
    TargetDir = 0;
  end;
  if (FWD_BTN && ~REV_BTN && ~STOP_BTN && ~BRK_FWD&& ~BRK_REV)
    TargetDir = 1;
  end;
  if (REV_BTN && ~FWD_BTN && ~STOP_BTN&&~BRK_FWD&& ~BRK_REV)
    TargetDir = -1;
  end;
```

PCB Design

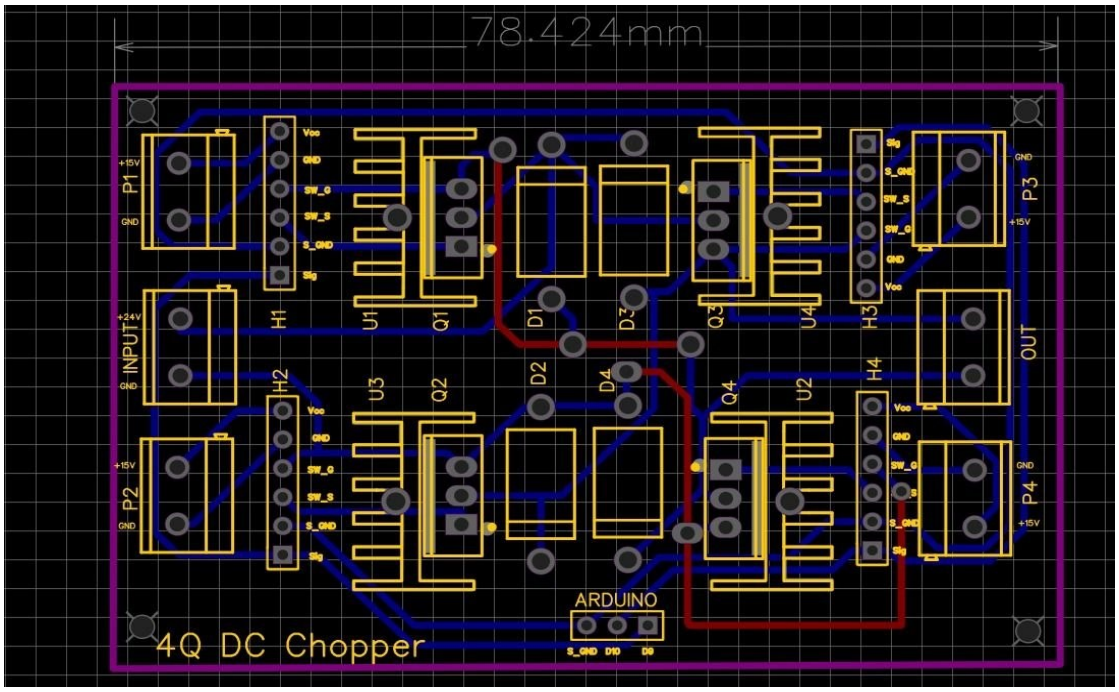
The PCB design was developed to implement the power and control circuitry of the DC motor drive system in a compact and reliable form. The design includes the MOSFET H-Bridge, TLP250 gate driver circuits, and necessary protection and filtering components.

Proper layout techniques were considered to minimize noise, reduce switching losses, and ensure safe separation between the low-voltage control section (ESP32) and the high-power stage. Wide copper tracks were used for power paths to handle high current levels, while proper grounding techniques were applied to improve system stability.

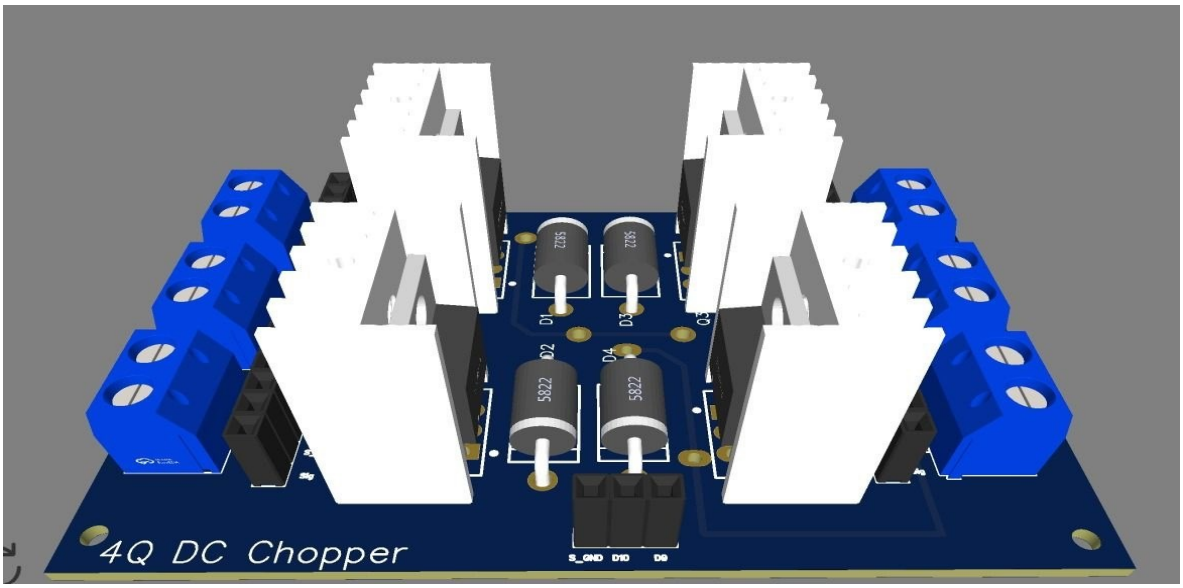
The final PCB design provides a practical and robust interface between the control unit and the motor drive stage.



Schematic



PCB Layout



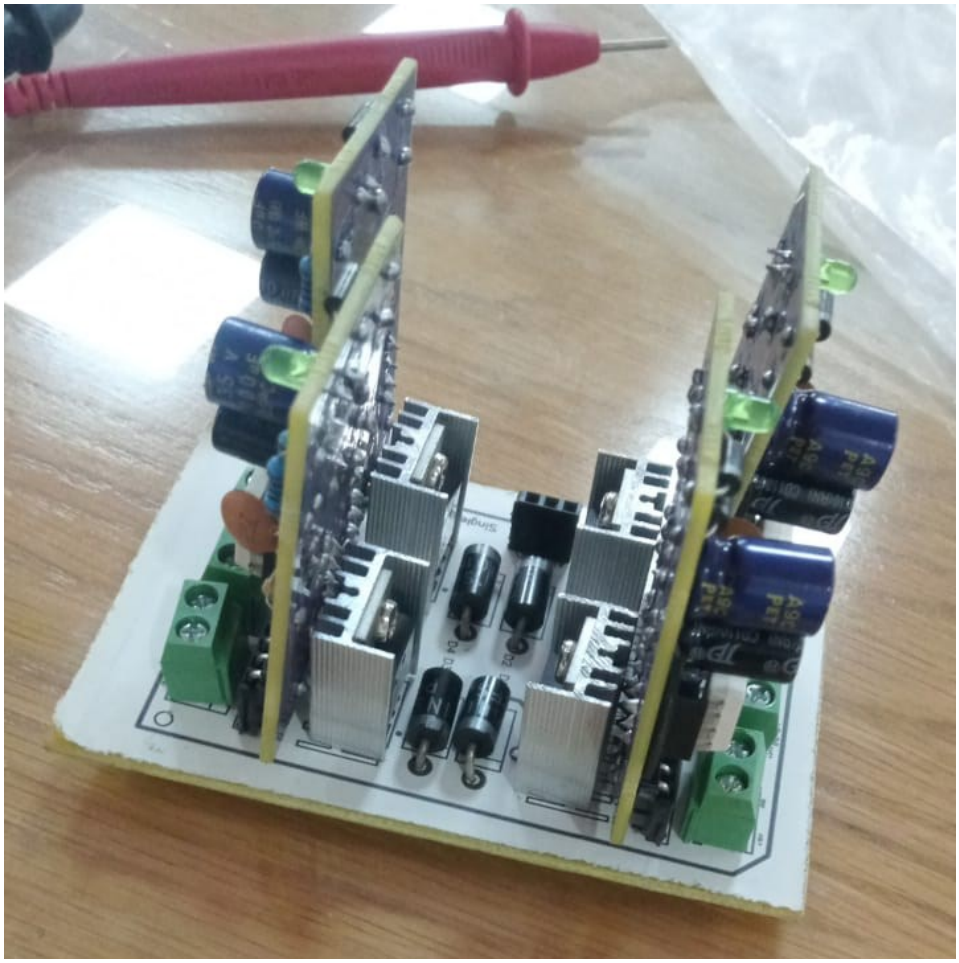
3D View

Practical Implementation (Real Hardware)

The practical implementation of the system was carried out using the designed hardware components, including the ESP32 controller, MOSFET H-Bridge, TLP250 gate drivers, and DC motor. The system was tested under different operating conditions such as forward rotation, reverse rotation, braking, and stop.

The control signals generated by the ESP32 were successfully used to drive the MOSFETs through the gate driver circuit. The system demonstrated stable operation, and the motor responded correctly to different input commands.

Safe switching between states was achieved using dead time implementation, which prevented short-circuit conditions in the H-Bridge. The experimental results confirmed the effectiveness of the proposed control strategy in real-time operation.



Conclusion

This project presented the design and implementation of an FSM-based DC motor drive system using the ESP32 microcontroller. The system successfully achieved control of motor speed and direction using Pulse Width Modulation (PWM) with an H-Bridge power stage.

Different operating modes, including forward rotation, reverse rotation, dynamic braking, and coasting, were implemented and managed using a Finite State Machine (FSM). This ensured safe and reliable transitions between states, with dead time applied to prevent shoot-through faults in the switching devices.

The use of TLP250 gate drivers provided proper isolation and reliable driving of the MOSFETs. The system was verified through simulation using MATLAB/Simulink and Proteus, and then implemented in hardware with PCB design.

The results demonstrate that the proposed system provides efficient, safe, and flexible control of a DC motor, making it suitable for industrial and educational applications.